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SMS spam detection using BERT and multi-graph convolutional networks

Linjie Shen [a], Yanbin Wang [b], Zhao Li [c], Wenrui Ma [a,d,*]

[a] School of Computer Science and Technology, Zhejiang Gongshang University, Hangzhou, Zhejiang, China
[b] Department of Engineering, Shenzhen MSU-BIT University, Shenzhen, 518172, Guangdong, China
[c] Zhejiang Lab, Hangzhou, Zhejiang, China
[d] Department of Big Data and Future E-Commerce Technology, Zhejiang Key Laboratory, Hangzhou, Zhejiang, China

A R T I C L E I N F O

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A B S T R A C T

The surge in smartphone usage has significantly increased Short Message Service (SMS) traffic and, consequently, SMS spam, posing risks such as phishing, financial losses, and privacy breaches. Traditional rule-based and blacklist methods fail against evolving spamming techniques, prompting the adoption of machine learning and deep learning approaches. However, models like Convolutional Neural Networks and Recurrent Neural Networks struggle to capture global co-occurrence patterns and complex semantics, while transformer-based models like Bidirectional Encoder Representations from Transformers (BERT) lack explicit syntactic and co-occurrence modeling. To address these limitations, we propose the BERT with Triple-Graph Convolutional Networks (BERT-G3CN) model, the first framework to integrate BERT word embeddings with graph embeddings from Co-occurrence, Heterogeneous, and Integrated Syntactic Graphs. This multigraph approach captures diverse features and models both global and local structures using tailored Graph Convolutional Networks. Experiments on two benchmark datasets demonstrate that BERT-G3CN achieves superior accuracy of 99.28 % and 93.78 %, representing an improvement of approximately 2–3 % over competitive baselines.

1. Introduction

The proliferation of smartphones has revolutionized the way people communicate [1], leading to an unprecedented surge in Short Message Service (SMS) usage worldwide. With billions of SMS messages sent daily, this communication medium has become integral to both personal and business interactions [2]. However, the massive increase in SMS traffic has also attracted the attention of malicious actors, resulting in a significant rise in SMS spam [3]. These unsolicited messages, which often contain phishing links and fraudulent offers, pose substantial risks to individuals and organizations alike.

The impact of SMS spam extends far beyond mere annoyance. Phishing attacks embedded in these messages can lead to severe financial losses, unauthorized access to sensitive information, and privacy breaches [4]. For businesses, the consequences can be even more dire, including damage to brand reputation, legal liabilities, and compromised customer trust. As spammers' tactics grow increasingly sophisticated, traditional filtering methods are often insufficient, leaving users vulnerable to the evolving threat landscape [4,5]. The sheer volume of SMS spam has created a pressing need for effective detection strategies [6]. Unlike email spam—which has received extensive research—SMS spam detection remains challenging due to the concise nature of messages and limited contextual information.

Given the financial and social implications of SMS spam, developing robust and efficient detection systems is urgent [7]. These systems must accurately identify spam, adapt to evolving spamming techniques, operate efficiently on resource-constrained devices, and scale to handle vast SMS traffic.

Historically, SMS spam detection began with rule-based systems and blacklists [8]. Although easy to implement, these methods quickly became ineffective as spammers adapted their tactics. Consequently, research shifted to machine learning (ML) algorithms [5], which offered greater flexibility by learning patterns directly from data. Early ML approaches like Naive Bayes, Support Vector Machines (SVM), and decision trees captured more complex data patterns [9], but frequently struggled with certain long-range dependencies and nuanced word interactions [10]. Over time, these limitations led to a decline in popularity for such methods in SMS spam detection.

The advent of deep learning (DL) marked a significant shift, with Convolutional Neural Networks (CNNs) and Recurrent Neural Networks

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* Corresponding author. School of Computer Science and Technology, Zhejiang Gongshang University, Hangzhou, Zhejiang, China
E-mail address: mawenrui@zjgsu.edu.cn (W. Ma).

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(RNNs) automatically learning hierarchical features from raw text data. These models captured more intricate patterns and dependencies, boosting spam detection performance. Yet, they still encountered difficulties in modeling broader contextual cues essential for nuanced interpretation [11]. More recently, transformer-based models like Bidirectional Encoder Representations from Transformers (BERT) have provided context-aware embeddings [12], as its self-attention mechanism effectively encodes word relationships. However, BERT may underperform when dealing with local dependencies and syntactic complexity in certain spam messages, potentially limiting accuracy.

Many approaches focus on word embeddings (e.g., GloVe, fastText, BERT) to capture semantic and contextual cues. However, such embeddings alone may overlook certain long-range or syntactic nuances in SMS messages. Meanwhile, Graph Convolutional Networks (GCNs) have been used in text classification to leverage global vocabulary relationships, often by constructing co-occurrence graphs. Yet, these methods tend to emphasize global word associations at the expense of local order or syntactic structure, which remain vital for accurate spam detection.

To address these limitations, we propose the BERT with Triple-Graph Convolutional Networks (BERT-G3CN) model for SMS spam detection, which integrates BERT-based word embeddings with graph embeddings derived from three distinct graph types: Co-occurrence Graphs, Heterogeneous Graphs, and Integrated Syntactic Graphs. The Co-occurrence Graphs focus on the frequency and patterns of word co-occurrence, enhancing the model's ability to detect recurring spam indicators. The Heterogeneous Graphs capture diverse semantic relationships by modeling interactions between different types of entities, such as parts of speech and named entities, thereby enriching the model's semantic understanding. Finally, the Integrated Syntactic Graphs incorporate syntactic dependencies to understand the grammatical structure, ensuring the model grasps the intended meaning and sentiment conveyed through word arrangement. By integrating these three types of graph embeddings, BERT-G3CN effectively captures both global relational patterns and local syntactic information, thereby overcoming the limitations of traditional GCN approaches and enhancing the accuracy and robustness of SMS spam detection.

Our approach provides a comprehensive understanding of the integrated message of a SMS message. For example, consider the SMS message: "Limited time offer available now."

- BERT Word Embeddings: BERT effectively captures the semantic meaning of each word, understanding "limited," "time," "offer," and "available" in context. However, it may not fully grasp "limited time offer" as a cohesive phrase indicating a spammy promotion.
- Co-occurrence Graphs: Identify that "limited," "time," and "offer" frequently co-occur in spam messages, reinforcing their association as indicators of spam.
- Heterogeneous Graphs: Model the relationship between "limited" (adjective) and "offer" (noun), enhancing the understanding that "limited time offer" is a specific promotional tactic.
- Integrated Syntactic Graphs: Capture the syntactic dependency between "limited time offer" and "available now," clarifying that the entire phrase forms a cohesive promotional message aimed at creating urgency.

These graph embeddings are processed through GCNs, which extract meaningful structural features from each graph type. By integrating the global relational information captured by the graphs with BERT's powerful contextual embeddings, our model achieves a comprehensive feature representation. This fusion not only improves the accuracy of SMS spam detection, but also offers a novel perspective on feature extraction, contributing to a deeper understanding and more effective approaches to SMS spam detection.

The main contributions of this paper are as follows.

1. **Triple Heterogeneous Graph Fusion.** We are the first to integrate Co-occurrence, part-of-speech (POS)/named-entity-recognition (NER-based) Heterogeneous, and Syntactic Dependency graphs for SMS spam detection, capturing both global word associations and local syntactic structure simultaneously. This novel fusion enhances the model's ability to understand diverse semantic cues and their interrelationships in the SMS domain.
2. **Cross-Layer Interaction between Local and Global Information.** Unlike previous methods that simply concatenate Transformer and GCNs representations, BERT-G3CN achieves a tight integration of local information from BERT and global semantic insights from graph embeddings. This is done by incorporating graph-based features into the Transformer layers, allowing local and global information to interact throughout the network and improve the final representation of the SMS content.
3. **SMS Spam Detection Focus.** This is the first application of a multi-graph GCNs combined with BERT for SMS spam detection. By leveraging the unique challenges of short, noisy, and evolving text,

BERT-G3CN achieves a 2–3 % improvement in accuracy over strong baselines, demonstrating the effectiveness of our approach in a real-world setting.

The remainder of this paper is structured as follows. Section 2 provides a comprehensive review of related works on SMS spam detection, covering the evolution of techniques from Rule- Based and Blacklist Methods to more advanced ML Approaches, DL Approaches, Transformer-based Models. In Section 3, we introduce the BERT-G3CN model, detailing its algorithm and design architecture. Section 4 describes the two datasets utilized in our experiments and presents the performance of BERT-G3CN on these datasets, comparing it with other related models. Finally in Section 5, we conclude this research based on our findings.

2. Related works

The field of SMS spam detection has seen considerable advancements over the years, evolving from simple rule-based systems to sophisticated machine learning and deep learning approaches. This section reviews the progression of SMS spam detection techniques and explains the novelty of our method accordingly.

2.1. Rule-based and blacklist methods

Early SMS spam detection systems relied heavily on rule-based approaches and blacklist mechanisms. These methods used predefined rules and maintained lists of known spam sources to filter out unwanted messages [13]. While easy to implement, these systems quickly became ineffective as spammers adapted their techniques to bypass the rules and blacklists. Consequently, the accuracy of such systems was limited, leading to a high rate of false positives and negatives.

2.2. Machine learning approaches

As the shortcomings of rule-based filters became evident, researchers turned to classical ML algorithms for SMS spam detection. One of the earliest and most widely adopted techniques in this domain is the SVM. Sjarif et al. [14] trained an SVM on the public SMS Spam Collection corpus, using a standard StringToWordVector pipeline followed by Information-Gain ranking to keep the top textual attributes. After stratified splits, their best model reached 98.9 % accuracy on the held-out set, outperforming Naive Bayes, Multinomial NB and several K-Nearest Neighbors (KNN) variants. Nevertheless, their bag-of-words representation ignores word order and relies on language-specific tokenization, limiting portability to multilingual or highly obfuscated spam.

Martínez-Mendoza et al. [15] later introduced the Smishing-4C corpus—the first open multi-class smishing dataset—and compared both traditional and transformer baselines. With a lightweight Bag-of-Words (BOW) vector augmented by six domain-specific features (length, spelling errors, phone, URL, slang, company name), a tuned Logistic-Regression classifier achieved an F1-score of 0.788, surpassing ERNIE (0.701) and BERT (0.700). Interestingly, concatenating the same 6-feature vector with BOW also allowed an SVM to reach 0.758 F1, confirming that well-engineered features can let linear models rival much heavier architectures while remaining far more efficient. Yet this approach still demands manual feature engineering and is evaluated on only 120 labeled samples, raising concerns about generalisability.

To push traditional ML further, Srinivasarao and Sharaff [7], proposed a hybrid Word2Vec + feature-selection pipeline. Word embeddings were first expanded via data augmentation, then pruned by six filter/wrapper selectors and an Equilibrium-Optimization meta-heuristic. The selected components were fed to a two-stage KNN–SVM ensemble whose parameters were tuned with Rat-Swarm Optimization, reaching 99.82 % accuracy on the SpamAssassin subset—a marginal gain over single SVMs but at the cost of a highly complex training workflow. Moreover, like most prior work, their evaluation remains binary and thus cannot capture the nuanced phishing categories emerging in modern SMS threats.

2.3. Deep learning approaches

As deep learning techniques matured, researchers moved beyond manual feature engineering to models that learn hierarchical representations directly from raw text. CNNs were among the first to show that a single-layer architecture with max-pooling can capture local n-gram patterns for short messages with little preprocessing [16]. Recurrent architectures soon followed: Sri et al. employed a Bidirectional Long Short-Term Memory (Bi-LSTM) with word-level embeddings and minimalistic preprocessing (case normalization, lemmatisation, stop-word removal), reaching 96.2 % accuracy on mixed email/SMS corpora and considerably reducing false positives relative to rule-based baselines [17].

Several recent works report that deeper or hybrid recurrent stacks outperform single-layer models when class imbalance is addressed. Kalyani et al. combined Bi-LSTM and GRU layers after ADASYN oversampling; their model attained 99 % accuracy on the popular Kaggle SMS dataset, surpassing both Random Forest (92 %) and plain LSTM variants [18]. In the Arabic-language setting, Kaddoura et al. fine-tuned a five-layer Bi-LSTM fed with multilingual GloVe vectors; despite the added morphological complexity, their best F1-score reached 97 % [19]. These results confirm that recurrent depth and robust sampling strategies can close the performance gap between classical ML and DL even on noisy, short texts.

Transformer-based encoders have pushed performance further by providing globally contextualized embeddings. BERT, pre-trained on large corpora and finetuned for spam or smishing classification, routinely exceeds 98 % accuracy while requiring little task-specific architecture [20,21]. Nevertheless, vanilla transformers still model tokens in isolation: they lack an explicit mechanism for representing syntactic dependency arcs or capturing higher-order co-occurrence patterns that signal imperative or promotional intent.

These limitations motivate our proposed solution. By integrating graph-based embeddings—which explicitly capture both global and local structural relationships—with BERT's powerful contextual representations, our BERT-G3CN model aims to overcome the above challenges and provide a more comprehensive feature extraction framework for SMS spam detection.

2.4. Graph convolutional networks approaches

Research on applying GCNs to SMS spam detection is relatively limited. However, inspiration can be drawn from analogous domains where GCNs have been successfully employed to identify spammers and detect spam content.

Zhiwei Guo et al. [22] proposed the Deep Graph Neural Network-based Spammer Detection (DeG-Spam) model, which first distinguishes stable and occasional relations in a heterogeneous social graph and then feeds the combined network to a deep Graph Neural Network. By explicitly modelling the latent, occasionally generated links via a parametric random-walk sampler, DeGSpam enriches the feature space and achieves roughly a 5–10 % performance gain over strong baselines on Twitter and Weibo datasets Although effective, the method is designed for user-interaction graphs and thus neglects the lexical and syntactic cues that are crucial in short SMS text.

Similarly, P. Jayashree et al. [23] introduced the Metapath-based GCN (M-GCN) framework for spam-review detection. M-GCN traverses predefined user–review–product metapaths in a heterogeneous network and attains about 96 % accuracy on Yelp and Amazon datasets, demonstrating the value of metapath-guided aggregation for spam detection in rich graph structures.

However, its reliance on hand-crafted metapaths limits flexibility when the domain, language, or spam tactics evolve.

Despite these advancements, current GCN-based models focus primarily on social or review graphs and therefore do not fully capture the diverse lexical relationships present in SMS messages. Moreover, approaches that depend on predefined metapaths can struggle to adapt to emerging spam patterns.

Addressing these challenges, the proposed BERT-G3CN integrates graph-based embeddings that encode global co-occurrence patterns, heterogeneous semantic relations, and syntactic dependencies alongside BERT's contextual word embeddings. This multi-faceted design ensures that both structural and lexical patterns are effectively captured, enhancing the model's ability to accurately classify SMS messages as spam or legitimate.

3. Methodology

3.1. Overview

The proposed BERT-G3CN model for SMS spam detection integrates BERT-based word embeddings with graph embeddings derived from three distinct graph types:

Co-occurrence Graphs, Heterogeneous Graphs, and Integrated Syntactic Graphs. This multi-graph approach captures diverse and complementary features, enabling the model to leverage both global and local textual patterns for enhanced spam detection. As shown in Fig. 1, the model first generates word embeddings from the input text using BERT, followed by constructing and processing the three graph types to capture co-occurrence relationships, semantic diversity, and syntactic dependencies. These graph embeddings are then fused with the word embeddings to form enriched representations, which are refined through self-attention layers before final classification, providing a holistic understanding of SMS content.

3.2. Graph construction

To illustrate how the proposed approach identifies features useful for detecting spam SMS messages, consider the example SMS message:

"Congratulations—you have won a gift; click the link to claim it now." A traditional BERT-based model may capture the contextual meaning of words but might struggle to generalize across structurally similar spam messages with different wording. The integration of graph-based embeddings provides additional insights:

(i) Co-occurrence Graphs capture frequent spam-triggering word pairs such as ("gift", "claim") and ("click", "link"), which often appear in fraudulent messages. (ii) Heterogeneous Graphs model linguistic properties, such as recognizing that "Congratulations": is commonly associated with promotional or deceptive messages when used in certain

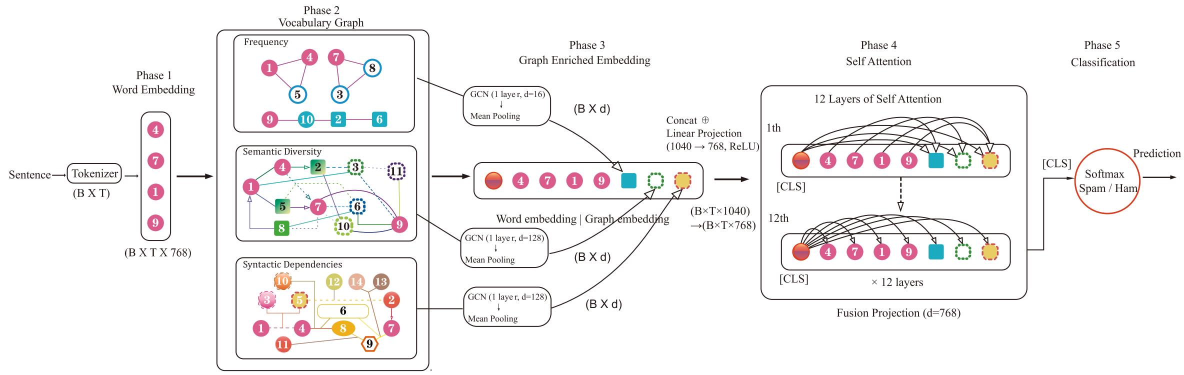


Fig. 1. Overview of the proposed BERT-G3CN model for SMS spam detection. **Phase 1** encodes the input SMS into token embeddings. **Phase 2** constructs three vocabulary graphs: the Frequency Graph, Semantic-Diversity Graph, and Syntactic-Dependencies Graph. **Phase 3** fuses graph-based embeddings with the original token embeddings. **Phase 4** applies a multi-layer Transformer (BERT) to refine the fused representations— arrows show multi-head attention across tokens. **Phase 5** classifies each SMS as spam or not using a fully connected layer on the [CLS] representation. Node and edge colors indicate distinct embeddings, nodes, or relationships, as shown in each sub-box.

contexts. (iii) Integrated Syntactic Graphs help capture imperative structures (e.g., "click the link" or "claim it now") that are frequently observed in SMS phishing attacks.

By leveraging such information, BERT-G3CN enhances its ability to generalize across different spam-message variations while minimizing false positives.

3.2.1. Co-occurrence graphs

Co-occurrence Graphs are constructed to capture the frequency and patterns of word co-occurrence within SMS messages. In this graph, each unique word in the vocabulary is represented as a node. Edges between nodes indicate that the connected words co-occur within a predefined window size w in the text. The weight of an edge between two words u and v is determined by the frequency of their co-occurrence.

Mathematically, the co-occurrence graph $G_{\text{co}} = (V, E_{\text{co}})$ is defined as:

$$\text{weight}(u, v) = \sum_{i=1}^{N} \sum_{j=i+1}^{i+w} I(w_i = u \text{ and } w_j = v) \tag{1}$$

where N is the total number of words in the SMS dataset, w_i and w_j are words at positions i and j respectively, and I is the indicator function that returns 1 if the condition is true, and 0 otherwise.

This graph effectively highlights global relational patterns by identifying frequently co-occurring word pairs, which can serve as strong indicators of spam. Unlike purely local contextual embeddings, these co-occurrence relationships are accumulated across all messages in the dataset, forming a global knowledge base that helps identify suspicious patterns regardless of minor wording variations.

3.2.2. Heterogeneous Graphs

Heterogeneous Graphs are designed to encode lexical, semantic, and named-entity information in a single multi-relational structure.

We instantiate the relation set

$$R = \{r_{wp}, r_{wn}, r_{ww}^{\text{sem}}\}, \tag{2}$$

and build each slice of the adjacency tensor $A_{\text{het}} \in R^{|R| \times |V| \times |V|}$ as follows:

Word–POS edges (r_{wp}). We run the spaCy en_core_web_sm tagger on every SMS.

For each token w labeled with part–of-speech tag p, we accumulate a global count

$$C_{wp}(w, p) := C_{wp}(w, p) + 1. \tag{3}$$

The weight is the corpus-level co-occurrence probability

$$\text{weight}_{wp}(w, p) = \frac{C_{wp}(w, p)}{\sum_{w', p'} C_{wp}(w', p')}. \tag{4}$$

Rare edges with $C_{wp} < 2$ are discarded to reduce noise.

Word–NER edges ($\mathbf{r_{wn}}$). Using the same spaCy pipeline, we detect named entities e (organization, URL, phone number, dots).

For every occurrence of token w inside entity e we increment

$$\mathrm{C_{wn}(w, e)}, \tag{5}$$

and normalize analogously to $\mathrm{r_{wp}}$.

Semantic-similarity edges ($\mathbf{r_{ww}^{sem}}$). Let $\mathrm{e_u, e_v}$ be the GoogleNews Word2Vec vectors of words $\mathrm{u, v}$.

An undirected edge is created if

$$\cos(e_u, e_v) > \theta_{\text{sem}} \tag{6}$$

with weight $\cos(\mathrm{e_u, e_v})$.

We set $\theta_{\text{sem}} = 0.7$, chosen via grid-search on the development set; 0.7 produced the best macro-F_1 while keeping the graph sparse.

All three normalized slices are finally stacked and symmetrized to form $\mathrm{A_{het}}$, which is consumed by the GCN encoder in Section 3.3.

3.2.3. Integrated Syntactic Graphs

Integrated Syntactic Graphs encode corpus–level dependency patterns in three clearly defined steps:

Dependency parsing. We parse every SMS with the spaCy en_core_web_sm parser, extracting directed arcs $(\mathrm{h} \rightarrow \mathrm{d})$ between head h and dependent d.

For each occurrence we accumulate a global count

$$C_{\text{syn}}[h, d] += 1. \tag{7}$$

Corpus-level aggregation and normalization. After parsing all messages we compute a symmetric frequency

$$C_{\text{syn}}^{\text{sym}}[u, v] = C_{\text{syn}}[u, v] + C_{\text{syn}}[v, u], \tag{8}$$

and convert it to a probability-like weight

$$\text{weight}_{\text{syn}}(u, v) = \frac{C_{\text{syn}}^{\text{sym}}[u, v]}{\sum_{u', v'} C_{\text{syn}}^{\text{sym}}[u', v']}. \tag{9}$$

Normalizing to $[0, 1]$ facilitates stable training of subsequent GCN layers.

Pruning. Edges with $\text{weight}_{\text{syn}}(u, v) < \delta$ are removed; we set $\boldsymbol{\delta = 10^{-4}}$. Grid search on the development set showed that this value achieves the best balance between sparsity and macro-F_1: smaller δ adds negligible recall but increases memory, while larger begins to discard informative long-tail relations.

The resulting undirected, weighted adjacency matrix A_{syn} retains highconfidence dependency links—e.g. the imperative pattern "click the link"—and feeds them to the GCN encoder.

Our motivation for choosing exactly these three graph types is twofold. First, they cover the three classical linguistic analysis levels—lexical, semantic, and syntactic—that prior work [24] has shown to be critical for text classification.

Second, our full ablation study (Section 4.7) empirically confirms that each graph contributes unique, non-redundant information: removing any single graph causes a 0.3–5 % drop in overall detection accuracy. Moreover, we experimented with adding further graph types (e.g. topic or discourse graphs) but observed only marginal performance gains at a substantial computational cost. These results demonstrate that the trio of Co-occurrence, Heterogeneous, and Syntactic graphs strikes the optimal balance between expressive power and efficiency for SMS spam detection.

3.3. Graph embedding through graph convolutional networks

Each graph branch uses one GCN layer to encode structural information into node representations. Let $\boldsymbol{H} \in R^{|V| \times 768}$ denote the initial node features, derived by mapping each graph node (vocabulary word) to its corresponding BERT embedding via a vocabulary index map. The adjacency matrix $\boldsymbol{A}$ is symmetrically normalized as $\widetilde{A} = D^{-\frac{1}{2}} A D^{-\frac{1}{2}}$. We update node features using:

$$H' = \mathrm{ReLU}(\widetilde{A}\, H\, W), \tag{10}$$

where $\boldsymbol{W}$ is a trainable weight matrix. Then, a graph-level vector is obtained by mean pooling:

$$g = \frac{1}{|V|} \sum_{v \in V} H'_v. \tag{11}$$

3.4. Integration with BERT word embeddings

After extracting graph embeddings from each of the three graph types using their respective GCNs, these embeddings are integrated with BERT-based word embeddings to form a comprehensive feature representation for each SMS message. By combining local contextual information from BERT with global graph-based representations, our model effectively learns both individual message characteristics and dataset-wide spam patterns, ensuring high accuracy in detecting spam across diverse linguistic variations.

3.4.1. BERT word embeddings

A pre-trained BERT model is utilized to generate contextual word embeddings H_{BERT}:

$$H_{\mathrm{BERT}} = \mathrm{BERT}(X) \tag{12}$$

where X represents the input SMS message tokens. BERT's self-attention mechanism captures the contextual relationships between words, providing rich semantic representations.

3.4.2. Feature fusion

The graph embeddings H_{co} , H_{het} , and H_{syn} obtained from the Co-occurrence, Heterogeneous, and Integrated Syntactic Graphs are concatenated with the BERT embeddings H_{BERT}.:

$$H_{\mathrm{fused}} = \mathrm{Concat}\left(H_{\mathrm{BERT}}, H_{\mathrm{co}}, H_{\mathrm{het}}, H_{\mathrm{syn}}\right) \tag{13}$$

To manage the increased dimensionality and ensure consistency, a linear projection is applied:

$$H_{\mathrm{proj}} = \mathrm{ReLU}\left(H_{\mathrm{fused}} W_{\mathrm{fuse}} + b_{\mathrm{fuse}}\right) \tag{14}$$

where W_{fuse} and b_{fuse} are learnable parameters. This projection reduces the dimensionality of the fused embeddings and introduces non-linearity, enhancing the feature representation.

3.4.3. Classification layer

The projected embeddings H_{proj} are then passed through a fully connected layer followed by a softmax activation function to perform the final classification:

$$\widehat{y} = \mathrm{Softmax}\left(H_{\mathrm{proj}} W_{\mathrm{cls}} + b_{\mathrm{cls}}\right) \tag{15}$$

where W_{cls} and b_{cls} are learnable parameters of the classification layer. The output $\widehat{y}$ represents the probability distribution over the classes (spam or legitimate).

4. Experimental results and discussions

In this section, we first describe the two SMS corpora and our experimental setup, including hyper-parameter choices and model configuration. We then compare BERT-G3CN's performance against a range of strong baselines, illustrate its convergence behavior via training and validation curves, and finally present ablation studies that quantify the contribution of each graph-based module.

4.1. Dataset

The datasets used in our experiments are gathered from two publicly accessible sources. The first dataset, collected from the UCI repository [25], consists of 5572 SMS messages (747 spam and 4825 ham) and is widely recognized as a standard benchmark in the SMS spam detection literature, despite being somewhat imbalanced (87 % ham vs 13 % spam). The second dataset, the ExAIS_SMS corpus [26], contains 4981 messages (2740 spam and 2241 ham) and offers a more balanced distribution (approximately 55 % spam vs 45 % ham). In this paper, we refer to these datasets as UCI_SMS and ExAIS_SMS, respectively. Table 1 presents key statistics of these two datasets, which together include diverse forms of unsolicited content and authentic messages, providing a realistic testbed for spam detection methods. Although other SMS spam datasets exist, many are proprietary or contain fewer samples, limiting reproducibility and direct comparisons with previous works. Accordingly, we focus on these well-known corpora to ensure a transparent and robust evaluation of our proposed approach, while also assessing the model's performance under both imbalanced and more balanced scenarios.

ExAIS_SMS allows us to assess model performance on a more balanced dataset, contrasting with the heavily skewed nature of UCI_SMS. This comparison provides a comprehensive evaluation of the model's robustness across different class distributions.

4.2. Configuration

All experiments were conducted in Python 3.10 using PyTorch together with the Hugging Face transformers library. To keep the search

Table 1
Comparison of UCI_SMS and ExAIS_SMS statistics.

Dataset	Label	Number of Instances	Class Distribution (%)
UCI_SMS	Ham	4825	86.59
	Spam	747	13.41
	Total	5572	100
ExAIS_SMS	Ham	2241	44.99
	Spam	2740	55.01
	Total	4981	100

space manageable while still covering the most influential knobs, we singled out three hyper-parameters—GCN depth, dropout, and learning rate—and tuned them with an explicit grid search (Tables 3–5). Concretely we tested GCN layers $\in \{1,2,3,4\}$, dropout rates $\in \{0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$ learning rates $\in \{1 \times 10^{-4}, 5 \times 10^{-5}, 2 \times 10^{-5}, 1 \times 10^{-5}, 5 \times 10^{-6}, 1 \times 10^{-6}\}$, while keeping the remaining settings fixed (see Table 2).

One GCN layer outperformed deeper stacks, confirming the classical over-smoothing effect on short texts: additional layers mixed node features so aggressively that graph information was diluted A dropout 0.5 achieved the best trade-off between under-fitting (rates ≤ 0.3) and over-regularisation (rates ≥ 0.6). For AdamW, a learning rate of 1×10^{-5} gave the most stable convergence; larger rates diverged, whereas smaller ones converged too slowly.

All remaining settings follow standard practice for efficient BERT fine-tuning. We keep the hidden size at 768 so that graph and token features share the same vector space without an extra projection. Graph outputs are projected to 16 dimensions for the dense but semantically narrow co-occurrence graph, and to 128 dimensions for the sparser yet richer heterogeneous and syntactic graphs, preserving their information with negligible memory overhead. A maximum sequence length of 128 covers more than 99 % of messages in both corpora, preventing truncation while keeping peak GPU usage below 6 GB. A batch size of 16 already saturates a single RTX 4090 (FP16); larger batches produced no further accuracy gain. Weight decay is fixed at (1×10^{-4}), the value recommended for BERT variants. Finally, training for ten epochs is sufficient for full convergence (see Fig. 3); additional epochs yield no measurable improvement.

4.3. Computational cost

Table 6 reports per-epoch training time, validation (inference) time, throughput, and peak GPU memory measured on a single NVIDIA RTX 4090 with batch size 16 and FP16 mixed precision. Graph construction is cached and therefore not included in the runtime.

As shown in Table 6, one training epoch finishes in approximately 5 min on a single RTX 4090, while validation (inference) processes around 12–14 s^{-1} using under 3.1 GB of GPU memory. In practice, BERT-G3CN is deployed on a gateway server rather than on end-user devices, ensuring that these computational demands do not affect the user's mobile experience.

4.4. Evaluation measures

To assess the effectiveness of our proposed BERT-G3CN model, we utilized several standard evaluation metrics: Accuracy, Precision, Recall, F1-Score, and AUC. These metrics provide a comprehensive view of the model's performance in classifying text.

Accuracy compares the predicted labels with the actual labels:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \tag{16}$$

where TP (True Positives) are the number of positive messages correctly identified as positive, TN (True Negatives) are the number of negative messages correctly identified as negative, FP (False Positives) are the number of negative messages incorrectly labeled as positive, and FN (False Negatives) are the number of positive messages incorrectly labeled as negative.

Precision evaluates the ability to avoid labeling negative messages as positive:

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{17}$$

Recall measures the ability to identify all positive messages:

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{18}$$

F1-Score is the harmonic mean of precision and recall:

$$\text{F1} - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{19}$$

Area Under the Curve (AUC) represents the model's ability to distinguish between classes, calculated from the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR):

$$\text{AUC} = \text{Area Under the ROC Curve}. \tag{20}$$

The use of these metrics ensures a detailed understanding of the classification capabilities of the BERT-G3CN model.

We utilized the sparse categorical cross-entropy as the loss function in our model because it is well-suited for handling classification tasks with multiple classes, especially when dealing with sparse data. This loss function is well-suited for classification tasks where the output is a probability distribution over multiple classes. The sparse categorical cross-entropy is defined as follows:

$$\text{Loss} = -\frac{1}{N}\sum_{i=1}^{N} \log p_i \tag{21}$$

where N is the number of samples and p_i is the predicted probability for

Table 2
Fixed training hyper-parameters used in all experiments.

Hyper-parameter	Value
Optimizer	AdamW
Weight decay	1×10^{-4}
Batch size	16
Number of epochs	10
Max sequence length	128

Table 3
Effect of GCN depth on BERT-G3CN accuracy on two benchmark datasets.

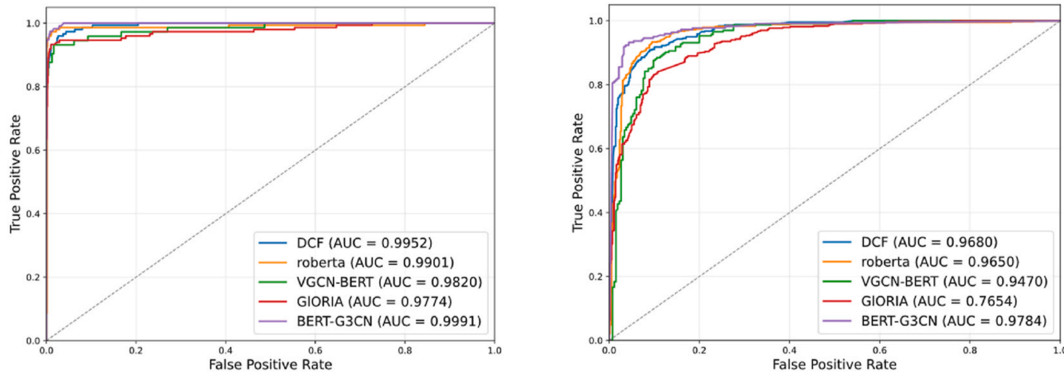
Depth	UCI_SMS Acc. (%)	ExAIS_SMS Acc. (%)
1	**99.28**	**93.78**
2	96.64	89.66
3	97.20	87.79
4	95.40	87.95

Table 4
Effect of dropout rate on BERT-G3CN accuracy on two benchmark datasets.

Dropout	UCI_SMS Acc. (%)	ExAIS_SMS Acc. (%)
0.2	97.76	92.75
0.3	95.62	89.46
0.4	98.67	88.85
0.5	**99.28**	**93.78**
0.6	97.52	91.78
0.7	96.78	90.09
0.8	86.64	86.73

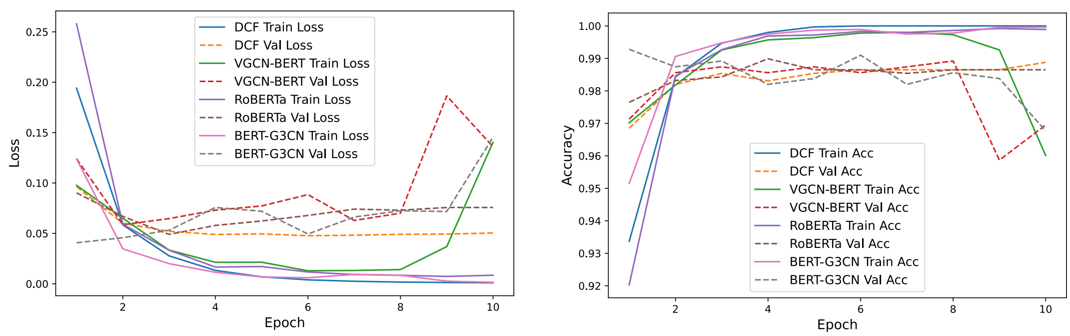
Table 5
Effect of learning rate on BERT-G3CN accuracy on two benchmark datasets.

Learning rate	UCI_SMS Acc. (%)	ExAIS_SMS Acc. (%)
1×10^{-4}	88.55	88.66
5×10^{-5}	98.39	87.73
2×10^{-5}	97.98	91.20
1×10^{-5}	**99.28**	**93.78**
5×10^{-6}	94.21	90.21
1×10^{-6}	96.53	85.65

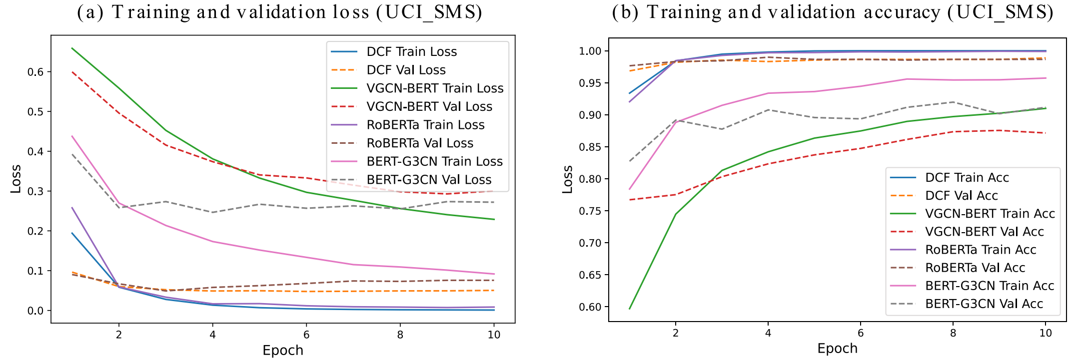


(a) (b)

Fig. 2. ROC Curves and AUC values for various models: (a) UCI_SMS, (b) ExAIS_SMS.



(a) Training and validation loss (UCI_SMS) (b) Training and validation accuracy (UCI_SMS)



(c) Training and validation loss (ExAIS_SMS) (d) Training and validation accuracy (ExAIS_SMS)

Fig. 3. Training progress of BERT-G3CN and baseline models.

Table 6
BERT-G3CN's runtime and memory usage.

Metric	UCI_SMS	ExAIS_SMS
Training time (s)	304.8	290.1
Training throughput (samples/s)	11.7	11.0
Training peak GPU mem (GB)	6.01	6.28
Validation time (s)	65.5	62.1
Validation throughput (samples/s)	13.6	12.8
Validation peak GPU mem (GB)	2.98	3.01

the true class of the $i-th$ sample.

The performance of BERT-G3CN, as measured by these metrics, is summarized in Table 7. The BERT-G3CN model demonstrates exceptional accuracy and robust performance across all metrics, setting a new standard in SMS spam detection tasks. This comprehensive evaluation experiment shows that BERT-G3CN not only achieves high accuracy but also excels in identifying both spam and ham messages, as reflected in its precision, recall, F1-score, and AUC values.

Table 7
Performance metrics of BERT-G3CN on UCI_SMS and ExAIS_SMS.

Dataset	Label	Precision	Recall	F1-score	Accuracy
UCI_SMS	Ham	0.9928	0.9990	0.9959	99.28 %
	Spam	0.9925	0.9500	0.9708	
ExAIS_SMS	Ham	0.9487	0.9384	0.9435	93.78 %
	Spam	0.9244	0.9369	0.9306	

4.5. Comparative study

To evaluate the performance of BERT-G3CN against stong and baselines, we compare it with DCF [27], RoBERTa [28], VGCN-BERT [29], and GLORIA [30]. The baseline methods are set with the same hyperparameters as reported in their original papers. DCF is an ensemble method that combines CNN-based feature extraction with random forests as classifiers; RoBERTa is built upon the BERT architecture with enhanced pre-training; GLORIA is a pure graph convolutional network; and VGCN-BERT integrates a word co-occurrence graph with BERT's self-attention to capture both local and global contextual cues from multiple perspectives. All experiments were run 10 times, and the results reported are the averages of these runs. Table 8 summarizes Precision, Recall, F1-score, and Accuracy on UCI_SMS and ExAIS_SMS.

As shown in Table 8, on the UCI_SMS, BERT-G3CN achieves the highest accuracy of 99.28 %, outperforming RoBERTa (98.92 %), VGCN-BERT (98.39 %), GLORIA (98.21 %) and DCF (98.02 %). When evaluated on the ExAIS_SMS, BERT-G3CN again leads with 93.78 % accuracy, followed by RoBERTa (92.28 %), DCF (90.76 %), VGCN-BERT (87.58 %) and GLORIA (86.86 %). Fig. 2 (a) and (b) illustrate the ROC curves and corresponding AUC values for both datasets. BERT-G3CN consistently attains the highest AUC, with RoBERTa, VGCN-BERT, GLORIA and DCF following in descending order. The superior performance of BERT-G3CN can be attributed to its integration of multiple GCN layers with BERT's self-attention, which captures both structural and semantic dependencies more effectively. Meanwhile, the strong results of RoBERTa and VGCN-BERT highlight the advantages of deep contextual embeddings and graph-augmented representations, respectively, in SMS spam detection tasks.

4.6. Training curves

To further illustrate the training dynamics of our BERT-G3CN model, we plot both the train-ing/validation loss and the training/validation accuracy over ten epochs for UCI_SMS and ExAIS_SMS in Fig. 3. For UCI_SMS, we observe that the training loss steadily decreases to a very low value, while the validation loss remains comparatively stable—indicating effective gen-eralization despite the strong class imbalance. Similarly, both training and validation accuracy improve rapidly, exceeding 90 % by around the fifth epoch On ExAIS_SMS, which is more balanced, the model continues to show stable convergence. Although the initial training loss and accuracy start at different levels compared to UCI_SMS, they quickly narrow the gap, demonstrating that BERT-G3CN can effectively learn discriminative representations in both skewed and relatively balanced class distributions. The curves for baseline models (DCF, VGCN-BERT, RoBERTa, and BERT-G3CN) are also included for comparison.

Notably, GLORIA is not included in the comparison as it requires 1000 epochs to converge, which exceeds the scope of this analysis. Overall, these curves highlight our model's robustness and its ability to maintain high performance under diverse data conditions.

4.7. Ablation study

To evaluate the impact of each graph-based module in BERT-G3CN, we conduct ablation experiments by systematically removing the Co-occurrence Graph, Heterogeneous Graph, or Integrated Syntactic Graphs. Table 9 presents the results of these experiments. The omission of the Co-occurrence Graph leads to a noticeable performance drop, especially on the more balanced ExAIS_SMS, where accuracy decreases to 87.96 %. Removing the Heterogeneous Graph maintains strong performance on UCI_SMS (99.19 % accuracy) but results in a decline on ExAIS_SMS (91.98 %), suggesting that heterogeneous features are particularly beneficial in balanced settings. Excluding the Integrated Syntactic Graphs module also reduces accuracy (92.58 % on ExAIS_SMS), emphasizing the role of syntactic cues in capturing sentence-level structures for distinguishing spam from legitimate messages. Overall, each module contributes to the model's effectiveness across different class distributions, demonstrating the complementary advantages of integrating co-occurrence, heterogeneous, and syntactic graph representations.

5. Conclusion

This paper presents the BERT with Triple-Graph Convolutional Networks (BERT-G3CN) model for SMS spam detection, integrating BERT-based contextual embeddings with graph representations from Co-occurrence, Heterogeneous, and Integrated Syntactic graphs. By leveraging Graph Convolutional Networks (GCNs) to extract structural relationships and combining them with BERT's powerful contextual encoding, BERT-G3CN effectively captures both global and local textual dependencies, addressing key limitations in traditional and deep-learning-based spam-detection approaches.

Experimental results show that BERT-G3CN outperforms all compared baselines across benchmark datasets, achieving accuracy rates of 99.28 % and 93.78 %, respectively. While these results highlight its effectiveness, real-world SMS spam detection presents continuous challenges, as spammers frequently adapt their tactics by introducing novel vocabularies, misspellings or obfuscations. BERT-G3CN's reliance on contextual and structural cues allows it to generalize well, but its performance could degrade if spammers craft adversarial messages mimicking legitimate text structures. To maintain robustness against evolving spam patterns, periodic model updates with fresh SMS data and hybrid detection strategies incorporating rule-based heuristics or anomaly detection are essential.

Looking ahead, we will focus on deployment-friendly adaptations of

Table 8
Comparison of performance metrics for different models on UCI_SMS and ExAIS_ SMS.

Dataset	Model	Label	Precision	Recall	F1-score	Accuracy
UCI_SMS	**VGCN-BERT**	Ham	0.9857	0.9959	0.9908	98.39 %
		Spam	0.9706	0.9041	0.9362	
	RoBERTa	Ham	0.9938	0.9938	0.9938	98.92 %
		Spam	0.9597	0.9597	0.9597	
	GLORIA	Ham	0.9866	0.9928	0.9897	98.21 %
		Spam	0.9510	0.9128	0.9315	
	DCF	Ham	0.9790	0.9979	0.9883	98.02 %
		Spam	0.9877	0.8889	0.9357	
	BERT-G3CN	Ham	0.9928	**0.9990**	**0.9959**	**99.28 %**
		Spam	**0.9925**	0.9500	**0.9708**	
ExAIS_SMS	**VGCN-BERT**	Ham	0.8773	0.8906	0.8839	87.58 %
		Spam	0.8739	0.8590	0.8664	
	RoBERTa	Ham	0.9321	0.9305	0.9313	92.28 %
		Spam	0.9108	0.9128	0.9118	
	GLORIA	Ham	0.8772	0.8913	0.8842	86.86 %
		Spam	0.8571	0.8394	0.8482	
	DCF	Ham	0.9073	0.9301	0.9186	90.76 %
		Spam	0.9080	0.8790	0.8933	
	BERT-G3CN	Ham	**0.9487**	**0.9384**	**0.9435**	**93.78 %**
		Spam	**0.9244**	**0.9369**	**0.9306**	

Table 9
Ablation study results on UCI_SMS and ExAIS SMS by removing different graph modules from BERT-G3CN.

Dataset	Variation	Label	Precision	Recall	F1-score	Accuracy	AUC
UCI_SMS	No Cooccurrence	Ham	0.9928	0.9948	0.9938	98.92 %	0.9953
		Spam	0.9660	0.9530	0.9595		
	No Heterogeneous	Ham	0.9918	0.9990	0.9954	99.19 %	0.9957
		Spam	0.9930	0.9463	0.9691		
	No ISG	Ham	0.9907	0.9959	0.9933	98.83 %	0.9932
		Spam	0.9722	0.9396	0.9556		
ExAIS_SMS	No Cooccurrence	Ham	0.9623	0.8182	0.8844	87.96 %	0.9699
		Spam	0.8038	0.9587	0.8745		
	No Heterogeneous	Ham	0.9333	0.9234	0.9283	91.98 %	0.9754
		Spam	0.9027	0.9151	0.9089		
	No ISG	Ham	0.9206	0.9501	0.9351	92.58 %	0.9763
		Spam	0.9330	0.8945	0.9133		

BERT-G3CN. Specifically, we plan to explore (i) knowledge-distillation and parameter-sharing schemes that compress the BERT backbone into lightweight teacher–student variants such as DistilBERT or TinyBERT; (ii) structured pruning and post-training 8-bit or 4-bit quantization techniques for both Transformer and GCN layers to reduce memory footprint; and (iii) graph sparsification methods or single-step spectral GCNs to alleviate neighborhood-aggregation overhead [31]. These efforts aim to transform BERT-G3CN from a high-accuracy research prototype into a practical, real-time SMS spam filter.

Overall, BERT-G3CN offers a promising and effective approach for mitigating SMS spam threats, enhancing security against phishing and fraudulent messages while demonstrating adaptability for real-world deployment.

CRediT authorship contribution statement

Linjie Shen: Writing – original draft. **Yanbin Wang:** Writing – review & editing. **Zhao Li:** Writing – review & editing. **Wenrui Ma:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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